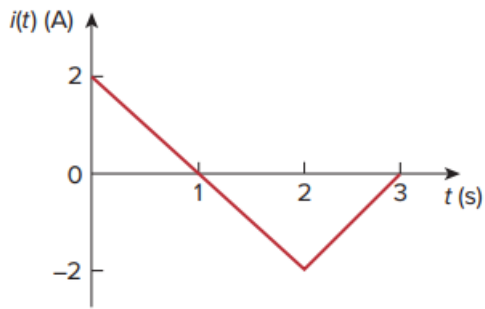


Problem

Refer to Fig. 7.39. Express $i(t)$ in terms of singularity functions.

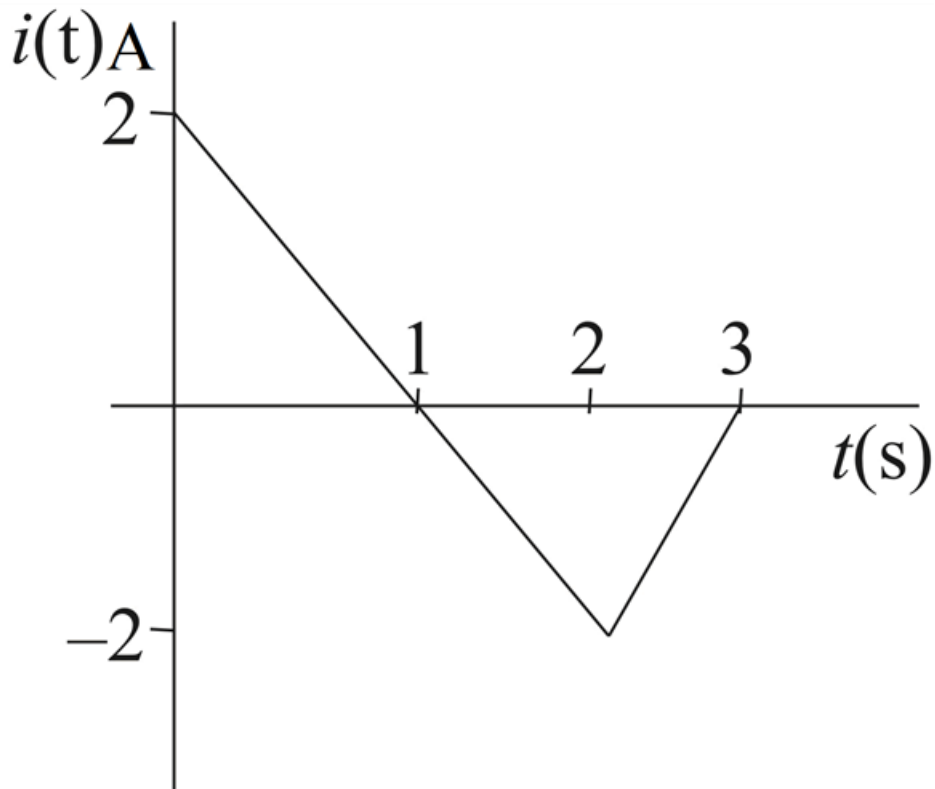
Fig. 7.39:



Step-by-step solution

Step 1 of 6

Given current signal is

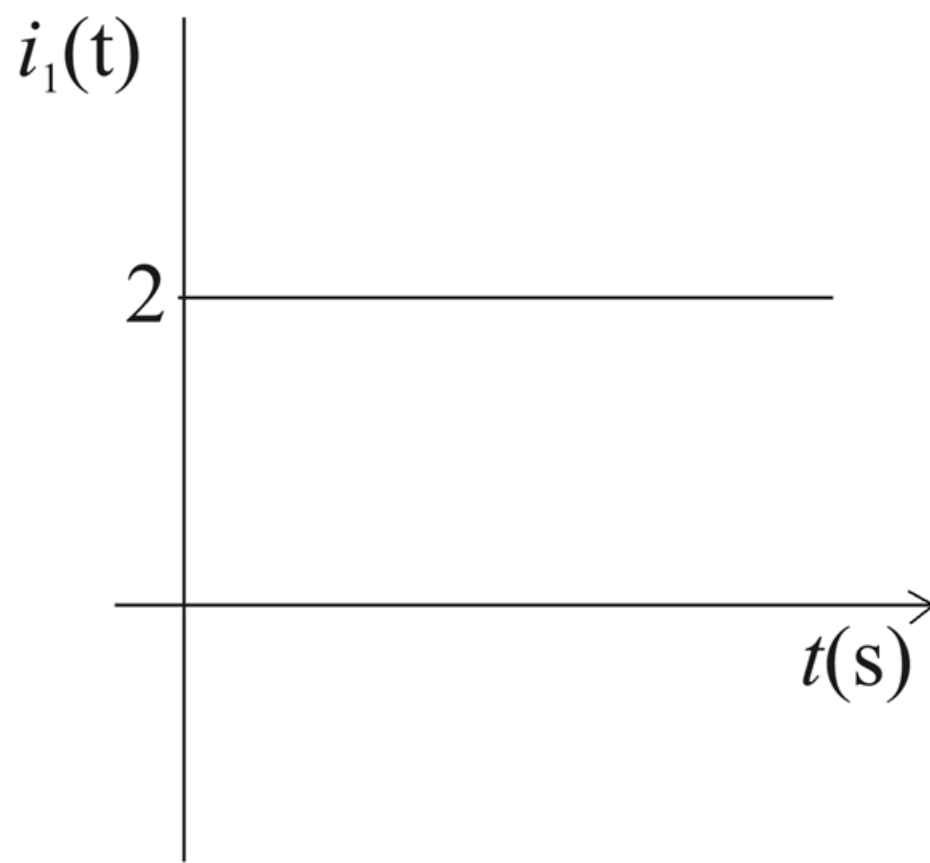


Step 2 of 6

Let $i(t) = i_1(t) + i_2(t) + \dots$

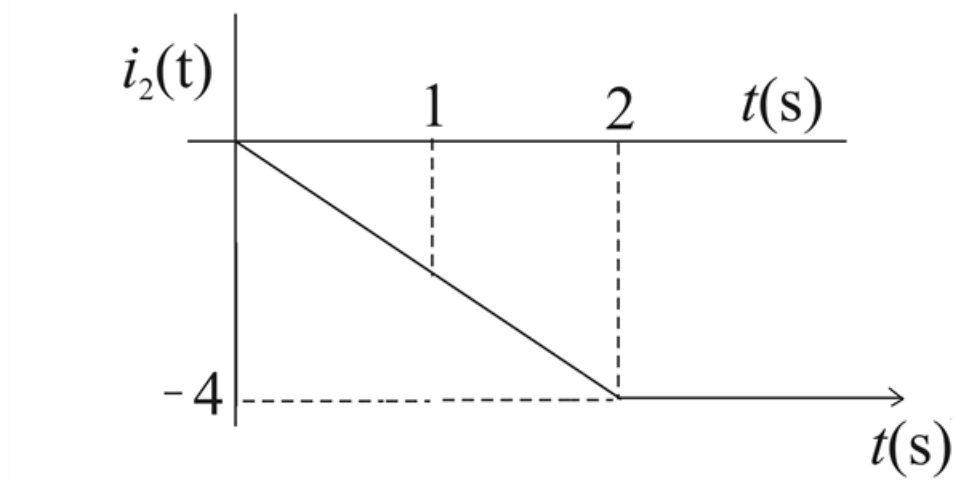
$i_1(t)$ is the step function with 2 units

Step 3 of 6



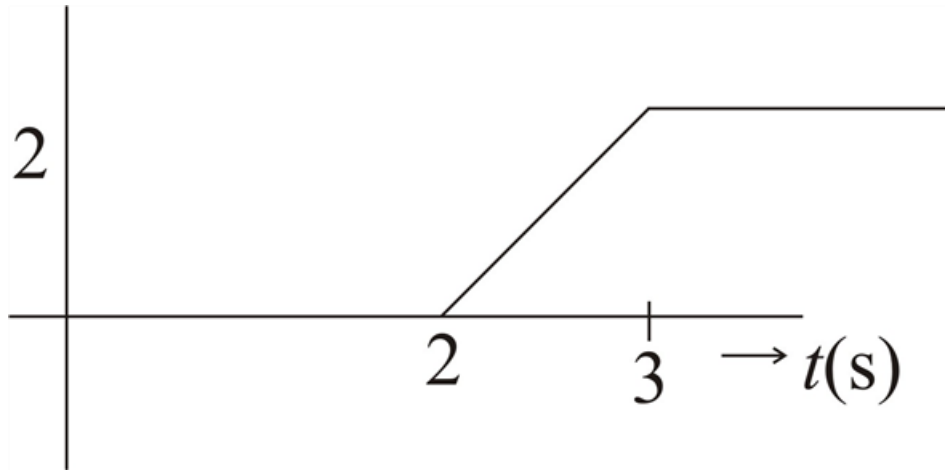
Step 4 of 6

And $i_2(t)$ is a ramp function with a slope of -2



Step 5 of 6

And take $i_3(t)$ is a ramp function with a slope of 2



Step 6 of 6

From the above three waveforms we can write

$$i_1(t) = 2u(t)$$

$$i_2(t) = -2(r(t) - r(t-2))$$

$$i_3(t) = 2(r(t-2) - r(t-3))$$

By combining as three plots we get the given wave form

Therefore

$$i(t) = i_1(t) + i_2(t) + i_3(t)$$

$$i(t) = 2u(t) - 2(r(t) - r(t-2)) + 2(r(t-2) - r(t-3))$$

$$i(t) = 2u(t) - 2r(t) + 2r(t-2) + 2r(t-2) - 2r(t-3)$$

$$\boxed{i(t) = 2u(t) - 2r(t) + 4r(t-2) - 2r(t-3)}$$